

BOL'SHOY SHANTAR, Russian Federation

WMO: 311740

Lat: **54.8623N**Lon: **137.4970E**Elev: **9**StdP: **101.22**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-31.5	-35.8	0.1	-33.2	-33.7	0.2	-30.9	12.5	-11.4	10.8	-12.7	2.1	30	0.789

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	21.1	16.0	19.1	15.0	17.6	14.1	17.0	19.8	15.8	18.2	14.7	16.8	3.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.8	11.2	18.3	14.7	10.4	17.1	13.7	9.8	16.0	48.1	20.0	44.5	18.3	41.5	16.9	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.2	7.8	-38.0	24.9	3.9	1.6	-40.8	26.0	-43.1	27.0	-45.2	27.8	-48.0	29.0		
(5)			-37.8	19.1	3.1	1.3	-40.0	20.0	-41.8	20.7	-43.6	21.4	-45.8	22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.2	-20.2	-19.5	-12.8	-4.2	2.4	7.2	11.6	12.8	8.6	1.1	-9.1	-17.3
	DBStd	12.45	5.11	5.08	5.23	3.95	2.68	3.37	3.05	2.54	3.10	3.66	5.68	5.44
	HDD10.0	5012	934	827	708	427	237	96	18	6	63	277	573	846
	HDD18.3	7862	1193	1060	966	677	495	333	211	173	292	535	823	1104
	CDD10.0	192	0	0	0	0	1	13	66	91	21	0	0	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	12	0	0	0	0	0	0	8	4	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	3.2	3.3	3.2	3.3	3.1	3.1	2.4	2.3	2.5	3.1	4.0	4.5	4.0
Precipitation	PrecAvg	524	24	17	16	35	48	46	63	69	69	65	40	35
	PrecMax	769	68	55	99	94	119	143	165	244	158	232	162	124
	PrecMin	204	0	0	0	2	7	8	5	0	4	6	0	1
	PrecStd	135	21	15	20	23	31	28	45	50	44	49	30	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.0	-3.6	2.5	7.3	15.6	21.1	24.0	23.4	19.8	12.8	4.2	-0.8
		MCWB	-3.3	-4.6	-0.7	2.9	8.7	14.6	17.9	17.7	14.6	7.9	1.6	-1.5
	2%	DB	-6.1	-6.4	-0.4	4.9	12.0	18.3	21.3	21.1	17.8	10.5	2.4	-2.8
		MCWB	-6.5	-7.8	-2.6	1.3	6.7	12.7	16.5	16.1	13.6	7.2	0.7	-3.4
	5%	DB	-9.7	-8.8	-2.2	3.1	9.5	15.9	18.8	19.3	16.4	8.8	0.8	-6.9
		MCWB	-10.5	-9.9	-3.9	0.3	5.3	11.5	15.4	15.3	12.5	6.0	-0.5	-7.7
	10%	DB	-11.8	-10.8	-3.7	1.5	7.3	13.6	16.8	17.8	15.0	7.3	-0.5	-9.5
		MCWB	-12.5	-11.7	-5.0	-0.5	4.1	10.5	14.3	14.6	11.7	4.6	-1.8	-10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.2	-4.1	-0.2	3.5	9.1	15.4	18.8	19.0	16.2	9.4	2.4	-1.3
		MCDB	-3.1	-3.7	1.9	6.2	14.0	19.6	22.8	22.3	18.4	11.1	3.5	-0.8
	2%	WB	-6.4	-7.6	-2.3	1.7	7.1	13.4	17.2	17.2	14.5	7.9	0.9	-3.6
		MCDB	-6.0	-6.7	-0.9	4.2	11.5	17.3	20.2	19.7	16.8	9.8	2.1	-2.9
	5%	WB	-10.5	-9.8	-3.7	0.6	5.7	11.9	15.7	16.1	13.3	6.3	-0.2	-7.6
		MCDB	-9.7	-8.9	-2.5	2.5	9.0	15.2	18.2	18.4	15.4	8.2	0.9	-7.0
	10%	WB	-12.5	-11.7	-5.0	-0.2	4.3	10.5	14.4	15.1	12.2	5.2	-1.7	-10.3
		MCDB	-11.8	-10.8	-3.6	1.3	6.9	13.4	16.5	17.0	14.3	6.9	-0.6	-9.7
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.9	12.7	9.2	7.1	9.1	8.1	8.7	10.3	7.9	6.6	7.9
		MCDBR	9.0	11.4	12.7	11.1	12.9	14.6	12.4	12.1	12.4	11.0	6.6	7.8
		MCWBR	8.7	10.5	10.7	7.9	8.2	9.4	8.2	7.7	8.3	8.0	5.9	7.6
	5% WB	MCDBR	8.8	11.2	11.1	9.3	11.7	13.5	11.3	10.5	10.4	9.7	6.1	7.8
		MCWBR	8.6	10.2	9.5	7.0	7.7	9.3	7.8	7.5	7.7	7.6	5.6	7.6
Clear-Sky Solar Irradiance	taub		0.261	0.286	0.348	0.457	0.518	0.503	0.516	0.445	0.382	0.371	0.320	0.275
	taud		2.044	2.018	1.868	1.648	1.671	1.934	2.016	2.243	2.360	2.174	2.106	2.018
	Ebn at Noon		650	773	793	746	727	751	731	769	777	664	561	525
	Edh at Noon		68	101	150	218	226	176	159	120	94	88	65	56
All-Sky Solar Radiation	RadAvg		0.95	2.00	3.43	4.43	4.42	5.15	4.58	3.98	3.11	1.77	0.90	0.62
	RadStd		0.07	0.17	0.36	0.44	0.84	0.57	0.62	0.52	0.29	0.19	0.11	0.09